

Engineer's Determination — Root Cause and Liability, Pier P2 Pier Cap Cracking

Nandira River Bridge Project — Package NRB-4

Document Reference:	ENG-NRB4-0083
Date:	06-May-2024
Contract No.:	NHIA/NRB4/CW/2020-01
From:	Engineer, Meridian Engineering Consultants
To:	Contractor, Sagara Constructions Pvt. Ltd.

Subject: Determination Under Sub-Clause 11.2 — Pier P2 Pier Cap Cracking

We refer to Defect Notice ENG-NRB4-0078 dated 20-Feb-2024, Crack Mapping Report CMR-NRB4-001, Laboratory Test Report LAB-NRB4-058, Contractor's Root Cause Analysis RCA-NRB4-004, and Minutes of Meeting MOM-NRB4-031, concerning the map cracking identified at the Pier P2 pier cap.

Having reviewed the particulars submitted, we make the following determination under Sub-Clause 11.2:

1. NATURE OF DEFECT: The cracking is confirmed, by Laboratory Test Report LAB-NRB4-058, to be shallow surface (plastic or early-age drying shrinkage) cracking, confined to approximately the top 20mm of the pier cap. The bulk concrete meets the specified characteristic strength (average corrected core strength 44.2 N/mm2 against 40 N/mm2 specified), and no alkali-silica reaction, sulfate attack, or reinforcement corrosion is present.
2. CAUSES EXCLUDED: We accept, on the evidence in LAB-NRB4-058 and RCA-NRB4-004, that this defect is not attributable to design, to the concrete mix or materials used, or to the ambient conditions prevailing at the time of the pour, each of which is excluded as a material contributing cause for the reasons set out in those reports.
3. CAUSE ACCEPTED: RCA-NRB4-004 records, and the Contractor did not dispute at the meeting recorded in MOM-NRB4-031, that the curing membrane and wet hessian cover for the exposed top surface of this pour were removed on 14-Aug-2021, approximately 3 days after placement, against the 7-day minimum curing period required by the Specification for exposed surfaces. We accept this finding, and consider it, on the balance of the evidence, the cause of the cracking: the shallow depth of the cracking (confined to the top 20mm) and its confinement to the exposed top surface only, with no equivalent cracking on the pier shaft or pier cap soffit, are each consistent with a curing-related cause and not with any of the causes excluded in paragraph 2.
4. LIABILITY: The failure to maintain curing protection for the period required by the Specification is workmanship not in accordance with the Contract within the meaning of Sub-Clause 11.2(b), and is not fair wear and tear. The Contractor is accordingly liable under Sub-Clause 11.2 for the cost of remedying this defect, and is not entitled to any additional payment or extension of time in respect of it.
5. NEXT STEPS: We will separately issue an instruction under Sub-Clause 11.1 for the rectification of the cracking, having regard to the Contractor's proposed method.

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ENGINEER'S DETERMINATION

Signed: Engineer's Representative
Meridian Engineering Consultants